

Day 47 - QoS (pt 2)

Wednesday, September 9, 2026

8:37 PM

QoS Info

QoS Classification

QoS's main purpose is to give certain kinds of network traffic priority over others during periods of congestion. This process is done through **classification** which organizes network traffic (packets) into traffic classes (categories).

- **Classification** is an essential part of **QoS**. To give priority to certain types of traffic, you have to identify which types of traffic to give priority to.
- **Classifying** traffic can be done through many methods, including:
 1. **ACL**
 - Traffic which is permitted by the **ACL** will receive certain treatment, while other traffic will not.
 2. **Network Based Application Recognition (NBAR)**
 - A method that performs **deep packet inspection**, looking beyond the layer 3 and 4 information and up to layer 7 to identify specific kinds of traffic.
 - In the layer 2 and layer 3 headers, there are specific fields used for this purpose.
 3. **Priority Code Point (PCP)**
 - It is a field in the **802.1Q** tag (in the **Ethernet** header) which can be used to identify low/high priority traffic.
 - It is only available when there's a **802.1Q** tag present.
 4. **Differentiated Services Code Point (DSCP)**
 - It is a field in the **IP** header, which can be used to identify low/high priority traffic.

Priority Code Point / Class of Service (PCP / CoS)

Priority Code Point (PCP) is also known as Class of Service (CoS). It is a field that is used in the 802.1Q tag to identify traffic priority.

- It has **3 bits** available which equals **8** possible values. ($2^3 = 8$)
- Because **PCP** is found in the **dot1q** header, it can only be used over:
 1. **Trunk links**
 2. **Access links with a voice VLAN**
- Each **PCP** value indicates a level of priority. These values are:

PCP value	Traffic Type
0	Best Effort (Default)
1	Background
2	Excellent Effort
3	Critical Applications
4	Video
5	Voice
6	Internetwork Control
7	Network Control

Some PCP value meanings

- 'Best Effort' delivery means there is no guarantee that data is delivered or that it meets any **QoS** standard. This is regular traffic.
- **IP Phones mark** call signaling traffic (which is used to establish calls) as **PCP 3**
 - They **mark** the actual voice traffic as **PCP 5**

IP Type of Service (ToS) Byte

The ToS byte (**8 bits**) consists of two fields which are used in modern networking. They are the DSCP and ECN fields of the IPv4 header.

- It previously used to be the **IPP** (**IP Precedence**) field, which consisted of **8 bits**, **5 bits** of which were reserved for various purposes and **3 bits** ($2^3 = 8$ values) were used for **classification**.
- It currently consists of the **DSCP** and **ECN** fields, where the **6 DSCP** bits ($2^6 = 64$ values) are used for **classification**.

IPP and DSCP Visualization

IP Precedence

Standard IPP markings are similar to PCP:

IPP value	Traffic Type
6 and 7	Reserved for 'network control' traffic. (ie. OSPF messages between routers)
5	Voice
4	Video
3	Voice Signaling
0	Best Effort

- With **6** and **7** being reserved, only **six** possible values remain.
- Although **six** values are sufficient for many networks, it wasn't enough for bigger and more complex networks. That's why it was replaced with **DSCP**.

Differentiated Services Code Point (DSCP)

After the change from IPP to DSCP, new standard markings had to be decided upon in order to simplify its implementation.

- The standard markings you need to know for the CCNA are:
 1. **Default Forwarding (DF)**
 - Best effort traffic.
 2. **Expedited Forwarding (EF)**
 - Low **loss/latency/jitter** traffic. (usually voice traffic)
 3. **Assured Forwarding (AF)**
 - A set of **12** standard values.
 4. **Class Selector (CS)**
 - A set of **8** standard values, provides backwards compatibility with **IPP**.

1. Default Forwarding (DF)

- **DF** is used for best effort traffic.
- The **DSCP** marking for **DF** is (**0**).

DF bits visualized

2. Expedited Forwarding (EF)

- **EF** is used for traffic that requires low **loss/latency/jitter** traffic. (usually voice traffic)
- The **DSCP** marking for **EF** is (**46**).

EF bits visualized

3. Assured Forwarding (AF)

- **AF** defines four traffic classes. All packets in a class have some priority.
 - The first (**3**) **bits** from the left represent the traffic class. (First number after the **AF** prefix)
 - The (**2**) **bits** after represent the **drop** precedence. (The second number after the **AF** prefix)
 - The very last bit at the end is a reserved bit which is always set to **0**.
- Within each, there are three levels of **drop** precedence.

- Higher **drop** precedence = more likely to **drop** the packet during congestion.
- The formula to convert from **AF** value to decimal **DSCP** value is:
 - $8X + 2Y$
 - Where **X** is the class number and **Y** is the drop precedence.

Example of an Assured Forwarding class (Class 4, drop precedence 3)
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3.1 AF Chart

Chart showing the different Assured Forwarding traffic classes and drop precedencies

4. Class Selector (CS)

Class Selector (CS) defines **eight** DSCP values for backwards compatibility with IPP

- The three extra **bits** that were added for **DSCP** are set to **0**, and the original **IPP bits** are used to make (**8**) values.
- The formula to convert from **CS** value to decimal **DSCP** value is:
 - $8X$
 - Where **X** is the class selector number.

IPP, CS and DSCP value conversions

RFC 4954 (QoS Standards)

RFC 4954 was developed to standardize QoS values and their uses.

- The most important recommendations are:
 - Voice traffic: **EF**
 - Interactive video: **AF4x**
 - Streaming video: **AF3x**
 - High priority data: **AF2x**
 - Best effort: **DF**

Trust Boundaries

The trust boundary of a network defines where devices trust and don't trust the QoS markings of received messages.

- If the markings are trusted, the device will forward the message without changing the markings.
- If the markings are untrusted, the device will change the marking according to the configured policy.
- If an **IP Phone** is connected to the switch port, it is recommended to move the **trust boundary** to the **IP Phones** by configuring the switch port connected to it.
 - In this case, the **IP Phone** is **trusted**, while the other end hosts are not.
 - If a user marks their PC's traffic with a high priority, the marking will be changed as it's **untrusted**.

Queueing/Congestion Management

[Review of QoS's queueing system](#)

- When a network device receives traffic at a faster rate than it can forward the traffic out of the appropriate interface, packets are placed in that interface's queue as they wait to be forwarded in a **FIFO** manner.
- When the queue becomes full, packets that don't fit in the queue are dropped (**Tail drop**) causing **TCP Global Synchronization**.
- **RED** and **WRED** drop packets early to avoid **tail drop**.
- An essential part of **QoS** is the use of multiple queues.
 - **QoS classification** plays a role in this, as the device can match traffic based on various factors (like the **DSCP** marking in the **IP** header) and then place it in the appropriate queue.
 - However, the device is only able to forward one frame out of a physical interface at once, so a **scheduler** is used to decide which queue traffic is forwarded next.

1. Queueing - Schedulers

- A common *scheduling* method is *weighted round-robin*.
 - *Round-robin* means that packets are taken from each queue in order, cyclically
 - *Weighted* means that more data is taken from high priority queues each time the *scheduler* reaches that queue.
- Another popular *scheduling* method is *Class-Based Weighted Fair Queuing (CBWFQ)*
 - It uses a *weighted round-robin scheduler* while guaranteeing each queue a certain percentage of the interface's *bandwidth* during congestion.

Round-robin scheduling is not ideal for voice/video traffic. Even if the voice/video traffic receives a guaranteed minimum amount of *bandwidth*, *round-robin* can add *delay* and *jitter* because even high priority queues have to wait for their turn in the *scheduler*.

CBWFQ Demonstration

2. Queuing - Low Latency Queuing (LLQ)

Low Latency Queuing (LLQ) designates one (or more) queues as *strict priority queues*. Which means that if there is traffic in the queue, the *scheduler* will always take the next packet from that queue until it is empty.

- *LLQ* is very effective for reducing the *delay* and *jitter* of voice/video traffic.
- However, it does have the downside of potentially starving other queues on the device if there is always traffic in the designated *strict priority queue*.
- *Policing* can control the amount of traffic allowed in the *strict priority queue* so that it can't take all of the link's *bandwidth*.

3. Queuing - Shaping and Policing

Traffic *shaping* and *policing* are both used to control the rate of traffic. In both cases, *classification* can be used to allow for different rates for different kinds of traffic.

1. Shaping

- It *buffers* traffic in a queue if the traffic rate goes over the configured rate.

2. Policing

- It *drops* traffic if the traffic rate goes over the configured rate
 - '*Burst*' traffic over the configured rate is allowed for a short period of time.
 - The amount of burst traffic allowed is configurable.